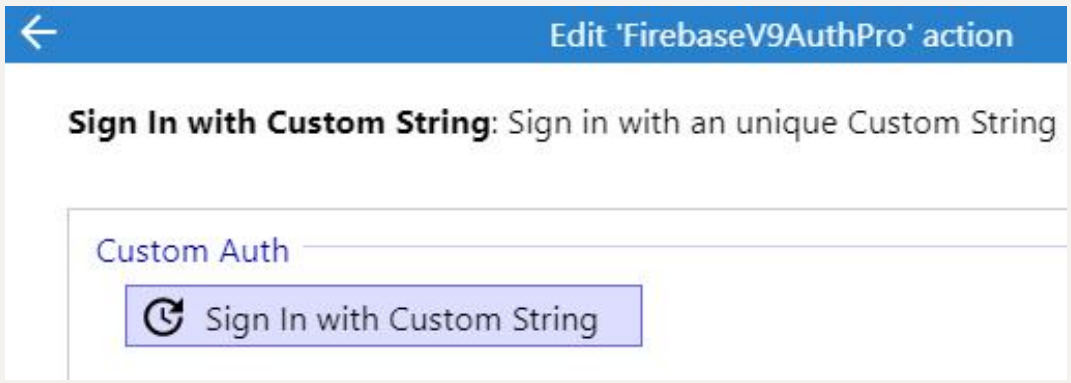
Auth: Sign In with Custom String

Last Updated: 13-Aug-2022

Introduction

Firebase Authentication PRO plugin allows to Authenticate an user using custom strings that must be unique to each other.



However, this action requires some extra steps to take before starting.
We need to host a node.js server and run some scripts there to make this work. To do this, we will use **Firebase Cloud Functions**. Firebase Cloud Functions is a service that allows you to host node.js apps on their server.

If you want to know about the Pricing of Cloud Functions, here are some links:
https://firebase.google.com/pricing
https://cloud.google.com/functions/pricing

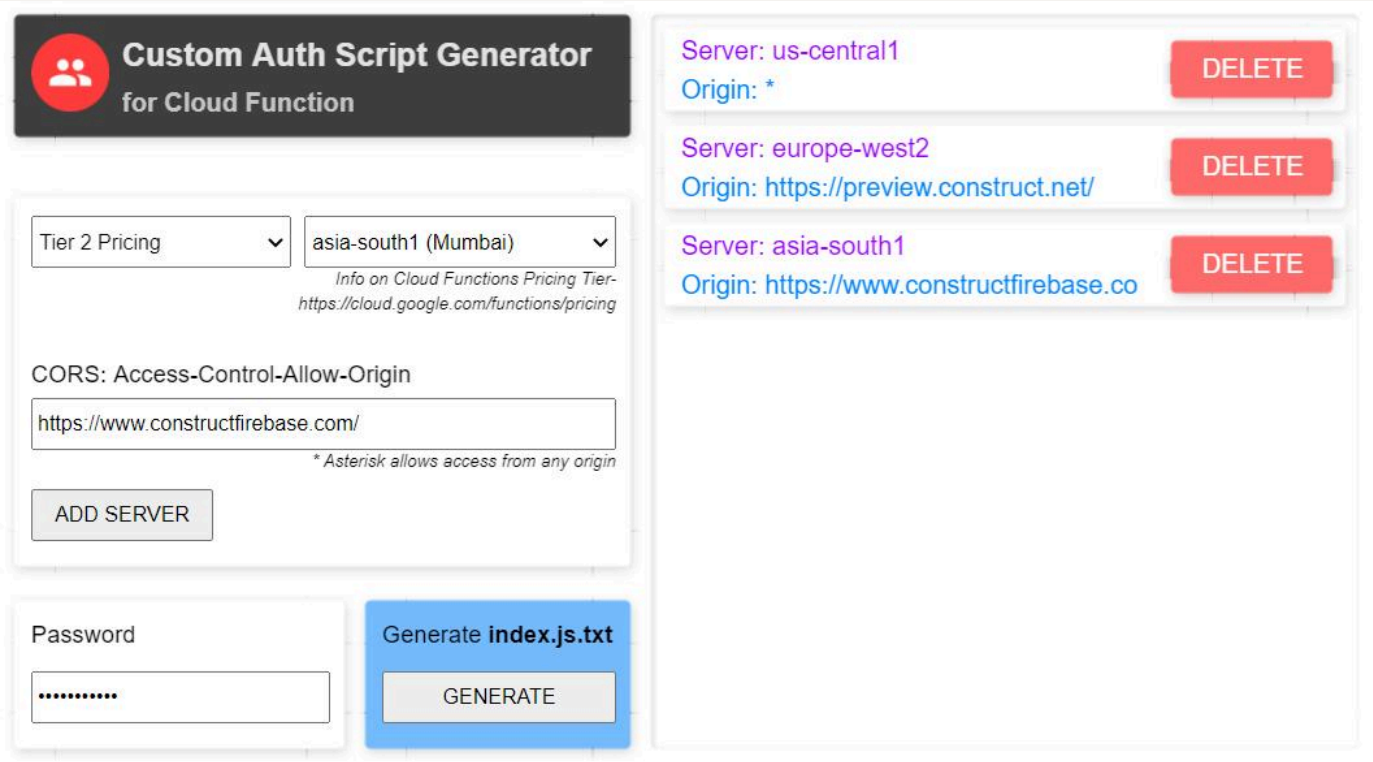
NOTE: You need the Firebase's BLAZE Plan to use Firebase Cloud Functions.

Step 1: Generate index.js server-side script

index.js contains the script that would run in Firebase Functions server. You can generate the content for this file according to your needs using the Custom Auth Script Generator Tool.

Link: **https://www.constructdatabase.com/tools/auth-pro-custom-script-generator/**

Access Pass: **authpropass01**



You can use this tool to choose your server location, and add multiple locations. Also it allows you to include an Access-Control-Allow-Origin for each server. (To know more about Origin: click here)
Enter a password, which will be used to authenticate user in server.
For testing purposes, just keep the default settings, i.e., Server: *us-central1* and Origin: *

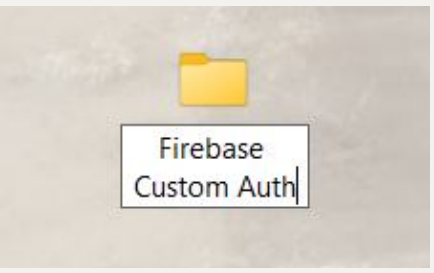
Finally, click Generate to download the ***index.js.txt*** file. You will need this file later in **Step 4**.

Step 2: Install Node.js

Download and install Node.js: https://nodejs.org/en/

Step 3: Install Firebase CLI & Initialize Project

To install Firebase CLI, we need Powershell (For Windows) or Terminal (for MacOS).



1. Create a folder where you want to work with your Firebase Project.

2. Open your Terminal in this folder. Type this command to navigate to this folder in Terminal:

```
cd “YOUR_LOCATION”
```

3. Now run these commands to install the required components:

```
npm install -g firebase-tools
npm install firebase-functions@latest firebase-admin@latest --save
```

4. Login to Firebase:

```
firebase login
```

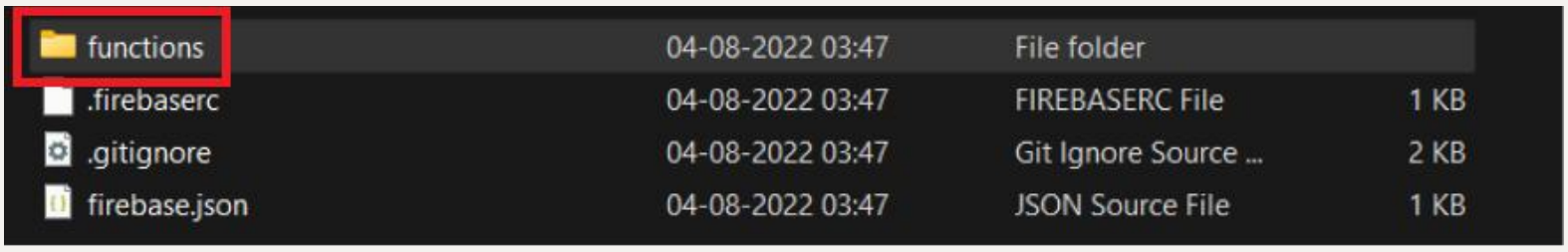
5. Initialize Firebase Functions.

```
firebase init functions
```

- Choose **JavaScript** when prompted “What language would you like to use to write Cloud Functions?”
- Press **N** when prompted “Do you want to use ESLint to catch probable bugs and enforce style?”
- Press **Y** when prompted “Do you want to install dependencies with npm now?”

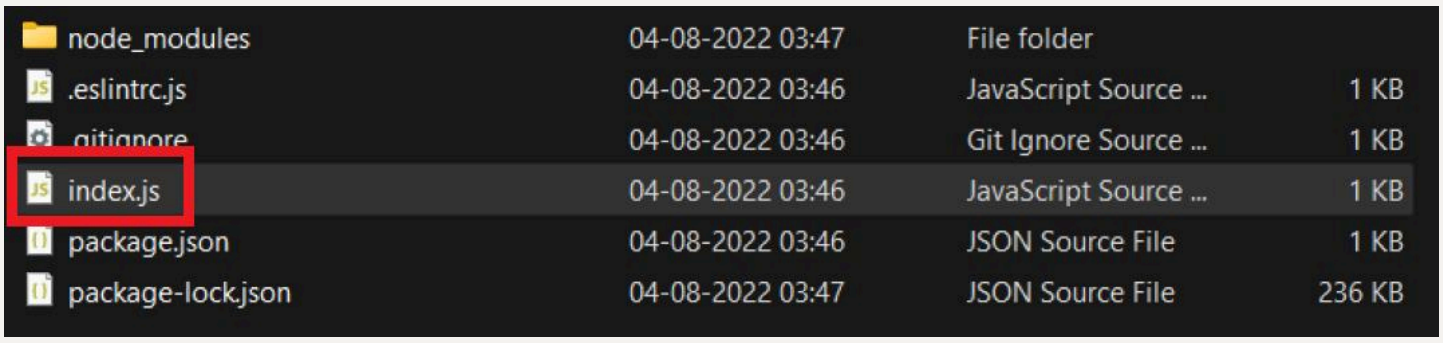
Step 4: Edit index.js file

1. Copy the content of ***index.js.txt*** that you generated in Setup 1



functions	04-08-2022 03:47	File folder	
.firebaserc	04-08-2022 03:47	FIREBASERC File	1 KB
.gitignore	04-08-2022 03:47	Git Ignore Source ...	2 KB
firebase.json	04-08-2022 03:47	JSON Source File	1 KB

2. In Firebase Functions project directory, Go to **functions** folder.

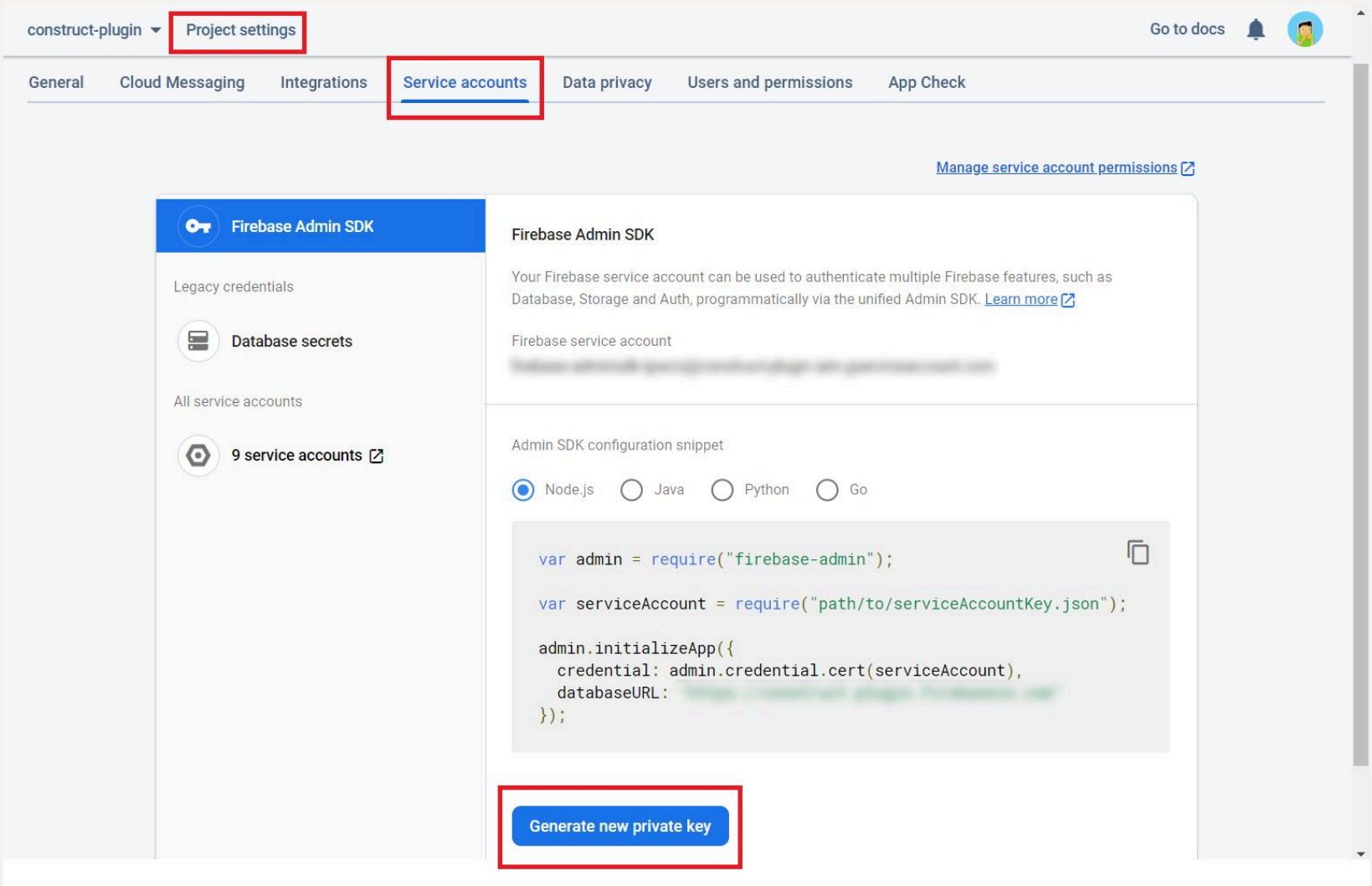


node_modules	04-08-2022 03:47	File folder	
.eslintrc.js	04-08-2022 03:46	JavaScript Source ...	1 KB
.gitignore	04-08-2022 03:46	Git Ignore Source ...	1 KB
index.js	04-08-2022 03:46	JavaScript Source ...	1 KB
package.json	04-08-2022 03:46	JSON Source File	1 KB
package-lock.json	04-08-2022 03:47	JSON Source File	236 KB

2. Open **index.js** file.

3. Clear the file and paste the contents of ***index.js.txt*** into this **index.js** file.

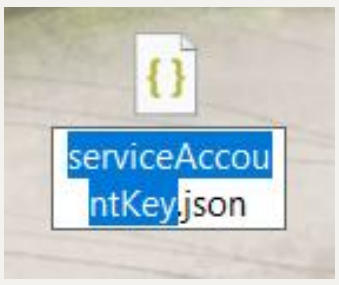
Step 5: Apply Service Account Key



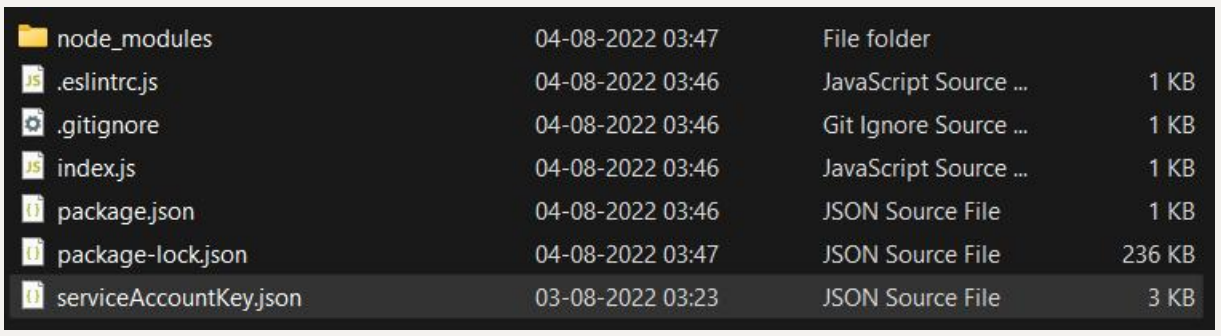
1. Go to your **Firebase Console > Project Settings**

2. Click **Service accounts** Tab

3. Click **Generate new private key**. A json file will be downloaded. Keep it confidential and never store it in a public repository.



4. Rename this file: **serviceAccountKey.json**



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serviceAccountKey.json	03-08-2022 03:23	JSON Source File	3 KB

5. Paste this file in your **functions** folder.

Step 6: Deploy to Firebase Function Server

Run this command to Deploy Firebase Functions to Server:

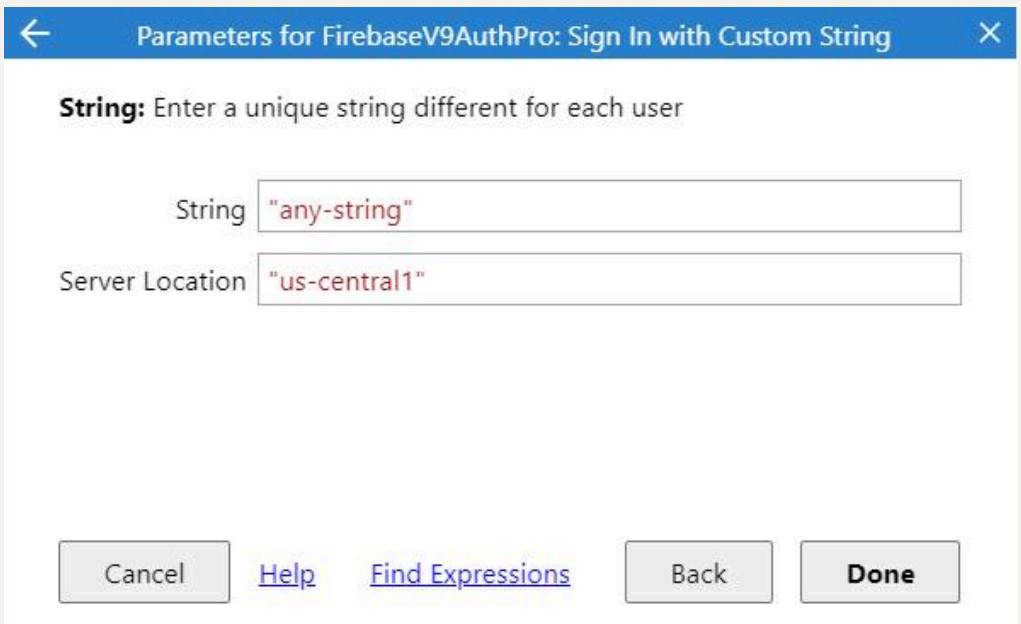
```
firebase deploy --only functions
```

Finally, you are now ready to go to your Construct 3 Editor.

Step 7: After Deployment

1. Go to the Google Cloud console: https://console.cloud.google.com/functions/
2. Click the checkbox next to the function to which you want to grant access.
3. Click **Permissions** at the top of the screen. The **Permissions** panel opens.
4. Click **Add principal**.
5. In the **New principals** field, type `allUsers`.
6. Select the role **Cloud Functions > Cloud Functions** Invoker from the Select a role drop-down menu.
7. Click Save.

Construct 3 Editor



String must be unique for each user.

You had selected Server Location while generating your index.js script in **Step 1**. Enter Server Location accordingly.